1. (习题七 38)

(1) 二者交点 $(0,0),(1,1)$, 则

$$\begin{aligned} S &= \left|\int_0^1 y\mathrm{d}y^2 - \int_0^1 x^2\mathrm{d}x\right| \\ &= \frac{1}{3}. \end{aligned}$$

(2) 从 $x_0 = 0$ 到 $x_1 = \frac{\pi}{4}$,

$$S_1 = \int_0^{\frac{\pi}{4}} (\cos x - \sin x)\mathrm{d}x = \sqrt{2} - 1.$$

从 $x_1 = \frac{\pi}{4}$ 到 $x_2 = \frac{5\pi}{4}$,

$$S_2 = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x)\mathrm{d}x = 2\sqrt{2}.$$

从 $x_2 = \frac{5\pi}{4}$ 到 $x_3 = 2\pi$,

$$S_3 = \int_{\frac{5\pi}{4}}^{2\pi} (\cos x - \sin x)\mathrm{d}x = 1 + \sqrt{2}.$$

则

$$S = S_1 + S_2 + S_3 = 4\sqrt{2}.$$

2. (习题七 40)

(1)

$$\begin{aligned} S &= \frac{1}{2}\left|\int_0^{2\pi} (x'y - y'x)\mathrm{d}t\right| \\ &= \frac{a^2}{2}\left|\int_0^{2\pi} ((1-\cos t)^2 - \sin t(t - \sin t))\mathrm{d}t\right| \\ &= \frac{a^2}{2}\left|\int_0^{2\pi} (2 - t\sin t - 2\cos t)\mathrm{d}t\right| \\ &= \frac{a^2}{2}|4\pi + 2\pi - 0| \\ &= 3\pi a^2. \end{aligned}$$

(3)

$$
\begin{aligned}
S &= \frac{1}{2}\left|\int_0^{2\pi}(x'y-y'x)\mathrm{d}t\right| \\
&= \frac{1}{2}\left|\int_0^{2\pi}\left(-\frac{3c^4}{ab}\cos^2 t\sin^4 t-\frac{3c^4}{ab}\sin^2 t\cos^4 t\right)\mathrm{d}t\right| \\
&= \frac{3c^4}{2|ab|}\left|\int_0^{2\pi}\sin^2 t\cos^2 t\mathrm{d}t\right| \\
&= \frac{3c^4}{2|ab|}\left|\int_0^{2\pi}2t(\sin t\cos^3 t-\cos t\sin^3 t)\mathrm{d}t\right| \\
&= \frac{3c^4}{4|ab|}\left|\int_0^{2\pi}t\sin 4t\cdot\mathrm{d}t\right| \\
&= \frac{3c^4}{8|ab|}.
\end{aligned}
$$

3. (习题七 42)

(2) 令 $x=r\cos t,\ y=r\sin t$, 则原方程给出

$$
r^2(\cos^4 t+\sin^4 t)=1\Rightarrow r^2=\frac{1}{\cos^4 t+\sin^4 t}\quad(t\in[0,2\pi]).
$$

因此

$$
\begin{aligned}
S &= \frac{1}{2}\int_0^{2\pi}\frac{\mathrm{d}t}{\cos^4 t+\sin^4 t} \\
&= \frac{1}{2}\int_0^{2\pi}\frac{\mathrm{d}t}{1-\frac{1}{2}\sin^2 2t} \\
&= 4\int_0^{\pi}\frac{\mathrm{d}t}{3-\cos t}.
\end{aligned}
$$

其中

$$
\begin{aligned}
\int\frac{\mathrm{d}t}{3-\cos t} &= \int\frac{\frac{2}{1+u^2}}{3-\frac{1-u^2}{1+u^2}}\mathrm{d}u \\
&= \int\frac{\mathrm{d}u}{1+2u^2} \\
&= \frac{1}{\sqrt{2}}\int\frac{\mathrm{d}(\sqrt{2}u)}{1+(\sqrt{2}u)^2} \\
&= \frac{\arctan\left(\sqrt{2}\tan\frac{t}{2}\right)}{\sqrt{2}}+C.
\end{aligned}
$$

则

$$
S=\sqrt{2}\pi.
$$

(4) 令 $x=r\cos t,\ y=r\sin t$, 则原方程给出

$$
r^2=\sin 2t\quad(t\in[0,\frac{\pi}{2}]\cup[\pi,\frac{3\pi}{2}]).
$$

所以

$$S = 2 \cdot \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin 2t \mathrm{d}t = 1.$$

4. (习题七 43)

绕 x 轴:

$$V_x = \pi \int_0^h (x^2)^2 \mathrm{d}x = \frac{\pi}{5} h^5.$$

绕 y 轴, 只需用 $x = h$ 绕成的圆柱减去 $x = \sqrt{y}$ 绕成的体, 则

$$V_y = \pi h^4 - \pi \int_0^{h^2} (\sqrt{y})^2 \mathrm{d}y = \frac{\pi}{2} h^4.$$

5. (习题七 46)

固定 $z \in [0, 1]$, 考虑 $S(\Omega_z)$, 初等几何分析可知, 若令 $\theta = \arcsin \sqrt{1 - z^2} = \arccos z$, 则

$$2S(\Omega_z) = \theta + (1 - z^2)(\pi - 2\theta) - \frac{1}{4} \sin 2\theta = \theta + (1 - z^2)(\pi - 2\theta) - z\sqrt{1 - z^2}.$$

研究各项不定积分:

$$\int \arccos z \cdot \mathrm{d}z = z \arccos z - \sqrt{1 - z^2} + C.$$

$$\int z\sqrt{1 - z^2} \mathrm{d}z = \frac{1}{3}(1 - z^2)^{3/2} + C.$$

$$\int z^2 \arccos z \cdot \mathrm{d}z = \frac{z^3}{3} \arccos z + \frac{1}{6} \int \frac{z^2 \mathrm{d}z^2}{\sqrt{1 - z^2}} = \frac{z^3}{3} \arccos z - \frac{1}{9}(1 - z^2)^{3/2} + \frac{1}{3}\sqrt{1 - z^2} + C.$$

代入下式即可.

$$\begin{aligned} V &= \int_0^1 2S(\Omega_z) \mathrm{d}z \\ &= -1 + \frac{2\pi}{3} + \frac{4}{9} - \frac{1}{3} \\ &= \frac{2\pi}{3} - \frac{8}{9}. \end{aligned}$$

> *Correction.*
>
> 听说用球坐标算方便一点. 回忆一下球坐标的 (r, θ, φ), 体积元是 $r^2 \sin\theta \cdot \mathrm{d}r \cdot \mathrm{d}\theta \cdot \mathrm{d}\varphi$ (注意三个切向恰好是正交的, 类似于二维下所谓的自然坐标架).

6. (习题七 48)

先考察对于极坐标上 $(0, 0), (a, \alpha), (b, \beta)$ $(a, b > 0,\ 0 \le \alpha < \beta \le \pi)$ 三点构成三角形绕极轴旋转的立体的体积. 初等几何分析得:

$$
\begin{aligned}
V_0 &= \frac{\pi}{3}b^3\sin^2\beta\cos\beta \\
&\quad + \frac{\pi}{3}(b^2\sin^2\beta + a^2\sin^2\alpha + ab\sin\beta\sin\alpha)(a\cos\alpha - b\cos\beta) \\
&\quad - \frac{\pi}{3}a^3\sin^2\alpha\cos\alpha \\
&= \frac{\pi}{3}a^2b\sin\alpha\sin(\beta-\alpha) + \frac{\pi}{3}ab^2\sin\beta\sin(\beta-\alpha) \\
&= \frac{\pi}{3}ab\sin(\beta-\alpha)(a\sin\alpha + b\sin\beta).
\end{aligned}
$$

若 $\beta = \alpha + \Delta\theta$, $\Delta\theta > 0$ 极小, 由于已有 $\sin(\beta-\alpha) = \Delta\theta + o(\Delta\theta)$, 其他因式的一阶小量可以忽略, 则令 $a = b = r$, 有

$$
V_0 = \frac{2\pi}{3}r^3\sin\theta\cdot\Delta\theta.
$$

因此, 对题目所求立体, 直接对 θ 积分, 有

$$
V = \int_\alpha^\beta \frac{\pi}{3}2r^3\sin\theta\mathrm{d}\theta = \frac{2\pi}{3}\int_\alpha^\beta r^3\sin\theta\cdot\mathrm{d}\theta.
$$

> *Correction.*
>
> 或者, 面积微元是 $r\cdot\mathrm{d}r\cdot\mathrm{d}\theta$, 重积分
>
> $$
> \int_\alpha^\beta \mathrm{d}\theta\int_0^{r(\theta)} r\mathrm{d}r\cdot 2\pi r\sin\theta
> $$
>
> 也能 (方便地?) 得到结果.

7. (习题七 49)

(2)

$$
\begin{aligned}
L &= \int_0^{2\pi}\sqrt{(a\theta)^2 + a^2}\mathrm{d}\theta \\
&= |a|\int_0^{2\pi}\sqrt{1+\theta^2}\mathrm{d}\theta.
\end{aligned}
$$

回忆有

$$
\begin{aligned}
\int\sqrt{1+x^2}\mathrm{d}x &= x\sqrt{x^2+1} - \int\frac{(x^2+1)-1}{\sqrt{1+x^2}}\mathrm{d}x \\
&= x\sqrt{x^2+1} - \int\sqrt{x^2+1}\mathrm{d}x + \int\frac{\mathrm{d}x}{\sqrt{1+x^2}} \\
&= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\int\frac{\mathrm{d}\sinh u}{\cosh u} \quad (x = \sinh u) \\
&= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}u + C \\
&= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\ln(x + \sqrt{1+x^2}) + C.
\end{aligned}
$$

所以

$$
L = \frac{|a|}{2}\left(2\pi\sqrt{1+4\pi^2} + \ln(2\pi + \sqrt{1+4\pi^2})\right).
$$

(4)

$$
\begin{aligned}
L &= a\int_0^{2\pi}\sqrt{(t\cos t)^2+(t\sin t)^2}\mathrm{d}t \\
&= 2a\pi^2.
\end{aligned}
$$